

Marina C. Kapitanov

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Motivated and creative second year biomedical engineering PhD student pursuing fellowship and research opportunities that build on communication and technical skills applying ultrasound imaging to women's health and extreme physiology.

EDUCATION

Lampe Joint Department of Biomedical Engineering at the University of North Carolina Chapel Hill (UNC) and North Carolina State University (NC State), Chapel Hill, North Carolina

PhD, Biomedical Engineering August 2025 - Present

Enhance Laboratory, Dr. Virginie Papadopoulos

Primary focus area: Ultrasound for fibroid embolization procedural monitoring

BS, Biomedical Engineering, German and Spanish minors August 2020 - May 2025

GPA: 3.596, *Magna Cum Laude*

Heidelberg University, Heidelberg, Germany

March 2022 - August 2022

University of North Carolina Baden-Wuerttemberg Consortium Participant

Nagoya University, Nagoya, Japan

September 2023 - August 2024

Nagoya University Program for Academic Exchange Participant

WORK EXPERIENCE

Graduate Research Assistant, Enhance Laboratory

August 2025 - Present

University of North Carolina at Chapel Hill, Chapel Hill, NC

Investigating the use of contrast enhanced ultrasound in uterine fibroid embolization procedures to improve clinical outcomes.

Undergraduate Research Assistant, Divers Alert Network,

January 2023 - Present

Divers Alert Network, Durham, NC

Analyzing salivary analytes and venous gas emboli scores from cardiac ultrasounds taken across 6 weeks of diving the same standardized dive profile amongst different divers in order to study individual variation in decompression stress and improve methods to avoid decompression sickness

AWARDS & FELLOWSHIPS

NIH T32EB034214 Predoctoral Training Grant

August 2025 - Present

Unified Medical Ultrasound Technology Development (UNMUTED) provides \$72,182 over two years and ultrasound technology curriculum, mentorship via a clinical and commercial mentor, clinical shadowing, research collaboration, commercialization workshops, and an internship in ultrasound research and development.

NC State Park Scholar

August 2020 - Present

A selective, full merit scholarship awarding over \$112,000 over eight semesters on the basis of scholarship, leadership, service, and character and involves participation in a four-year Enrichment Academy including seminars with distinguished speakers, executive-style training, and exploring leadership challenges regionally and nationally.

PROFESSIONAL ACTIVITIES & MEMBERSHIP

NSF Regional I-Corp Participant (Customer Discovery & Market Research)

Summer 2026

Institute of Electrical and Electronics Engineers (IEEE) Student Member

Summer 2026 - Present

Undersea and Hyperbaric Medicine Society Member

Summer 2024 - Present

Engineers Without Borders: Research Triangle Park Professional Chapter Member

Summer 2024 - Present

POSTERS AND PUBLICATIONS

Marina C. Kapitanov, Erin T. Carey, Maureen P. Kohi, Virginie Papadopoulos. "Contrast-enhanced ultrasound use in uterine artery embolization procedure: A scoping review." UNC Radiology Symposium 2026 (poster and abstract).

Joshua B. Currens, Rhiannon J. Brenner, Madeline Coombs, Katherine Eltz, Danica Grant, **Marina C. Kapitanov**, Virginie Papadopoulos, Frauke Tillmans. "Divers tend to produce individually consistent venous gas emboli grades after conducting the same dive multiple times" Undersea & Hyperbaric Medical Society Annual Scientific Meeting 2025 (oral, poster and abstract).

Berger, M. B., Cohen, D. J., Bosh, K. B., **Kapitanov, M.**, Slosar, P. J., Levit, M. M., Gallagher, M., Rawlinson, J. J., Schwartz, Z., & Boyan, B. D. (2023). Bone marrow stromal cells generate an osteoinductive microenvironment when cultured on titanium-aluminum-vanadium substrates with biomimetic multiscale surface roughness. *Biomedical materials* (Bristol, England), 18(3), 035001.